

Sales Intelligence Agent

Build Guide

How to build your own AI-powered sales research, target scoring & meeting prep tool using Claude Cowork.

Who this is for

Any sales team that wants to research prospects faster, score them consistently, and walk into every meeting prepared. Originally built for a partnership BD team, fully adaptable to any industry or sales motion.

What you get

A working web app with prospect research, AI-generated briefs, fit scoring, outreach pipeline, activity log & PDF export. Runs on Vercel. Powered by the Anthropic API.

Prerequisites

- GitHub account (free)
- Vercel account (free Hobby tier works)
- Anthropic API key (pay-per-use, ~\$5–20/mo typical)
- Node.js 18+ installed locally
- Basic comfort with a terminal
- Claude Cowork (for day-to-day use)

1. What the Tool Does

The Sales Intelligence Agent is a custom web application that acts as a research and meeting prep co-pilot for your sales team. It combines AI reasoning (via the Anthropic API) with your own sales context to produce structured, actionable intelligence on any prospect or partner target — in seconds.

Feature	What it does
Prospect Brief	Enter any org name and get a full AI-generated profile: industry, size, fit score, value props, pitch angles, etc.
Fit Scoring	Every org receives a 0–100 score based on criteria you define (signals, industry match, network size, travel frequency, etc.)
Outreach Pipeline	A workflow panel to batch-generate personalised cold emails, manage follow-up stages and track deal progress.
Activity Log / Reports	A running log of every contact touched, email sent, stage change and follow-up due date. Exportable.
Saved Briefs	Researched orgs are saved in the browser so you can pull them up before any call without re-running research.
PDF Export	One-click export of any brief as a clean, branded PDF to share with colleagues before a meeting.

2. Architecture Overview

The tool is a Next.js web application deployed on Vercel. The frontend is a single React page. The backend consists of serverless API routes that call the Anthropic API. There is no database required to get started — browser localStorage handles brief persistence. Optional integrations (ZoomInfo, Crossbeam) slot in alongside the core Anthropic call.

Layer	Technology	Role
Frontend	Next.js 16 + React 19	Single-page app — UI, state, navigation
Hosting	Vercel (Hobby = free)	Deploys automatically from GitHub on every push
AI Engine	Anthropic API (claude-haiku-4-5)	Runs prospect research, email generation, scoring and synthesis
Auth / Storage	localStorage (default) or Supabase (optional)	Saves briefs, style profiles, report entries
Enrichment	ZoomInfo MCP (optional)	Adds verified employee count, revenue, HQ data to briefs
Overlap Intel	Crossbeam API (optional)	Flags existing partner relationships as warm intro signals

The Vercel Hobby plan's 10-second serverless function limit is the key constraint to be aware of. The partner-research route uses Edge Runtime (30s limit) to work around this. Upgrading to Vercel Pro removes this

constraint entirely.

3. Step-by-Step Setup

Step 1 — Fork or clone the repository

The codebase is a standard Next.js project. Start by getting a copy into your own GitHub account:

- Go to github.com and create a new repository, or fork an existing Next.js boilerplate
- Clone it to your local machine: `git clone https://github.com/your-org/your-repo.git`
- Install dependencies: `npm install`
- Run locally to verify: `npm run dev` — you should see the app at `localhost:3000`

Step 2 — Get your Anthropic API key

- Go to console.anthropic.com and create an account
- Navigate to API Keys and create a new key
- Keep it safe — you will add it to Vercel's environment variables, not to your code
- Recommended model: `claude-haiku-4-5-20251001` (fast, affordable, strong reasoning)
- Typical cost: \$5–20/month for an active sales team. Set a spend limit in the console to be safe.

Step 3 — Deploy to Vercel

- Go to vercel.com, create a free account and connect your GitHub
- Import your repository — Vercel auto-detects Next.js and configures the build
- In Project Settings → Environment Variables, add: `ANTHROPIC_API_KEY` = your key
- Deploy. Your app will be live at `your-project.vercel.app` within ~2 minutes
- Every git push to main auto-deploys — no manual steps needed after initial setup

Step 4 — Set up your Style Profile

The first time you open the app, you'll be prompted to set up a style profile. This is how Claude learns to write emails in your voice and understand your target segment. Paste in 2–3 examples of emails you've sent, and describe your focus area (e.g. 'mid-market sports tech teams, \$10M–\$100M ARR, VP Sales and RevOps personas').

Step 5 — Customise the scoring prompt (critical)

Open `app/api/partner-research/route.ts`. The scoring criteria and partner framework live in the prompt string. This is the single most important thing to customise — it controls what kinds of orgs score high and what signals the AI looks for. See Section 4 for guidance on how to adapt this.

4. Customising the Scoring for Your Sales Motion

The fit score (0–100) is generated by Claude based on criteria you define in the prompt. The current scoring framework was built for a partnership BD team focused on SMERF organisations. For a sports tech sales team, you would replace that framework with your own ICP criteria.

The scoring prompt structure

Each scoring rule is a line in the prompt that tells Claude: 'if you find this signal, add N points'. Example:

```
fitScore 0-100: +20 Has a dedicated sports operations or travel team +15 Runs recurring events (tournaments, leagues, training camps) +15 ICP industry match (pro sports, college athletics, sports tech, esports) +15 Uses manual/spreadsheet-based ops (high switching potential) +10 Multi-team or franchise network (wide distribution) +10 100+ full-time staff +5 Currently evaluating or recently switched tech vendors -15 Consumer-only focus with no B2B operations -10 Single event or highly seasonal operation only
```

'What makes a good target' framework

Before the scoring rules, describe your ideal customer profile as 4–5 qualitative criteria. Claude uses these to reason holistically before assigning the score:

```
Score this org as a sales target using these criteria: 1. OPERATIONAL COMPLEXITY – manages logistics across multiple teams, venues or events 2. TECHNOLOGY GAP – currently doing things manually that your product automates 3. GROWTH TRAJECTORY – expanding roster, new markets, or growing event calendar 4. BUYING POWER – budget authority exists at the VP or Director level 5. REFERENCEABLE FIT – a win here would be a strong case study for your segment
```

Tier thresholds

Set your tiers to match how your team thinks about pipeline quality:

- Hot / Strong ≥ 70 — high-confidence ICP, fast-track to outreach
- Warm / Potential ≥ 40 — worth qualifying, needs more discovery
- Cold / Low < 40 — not a fit now, may revisit later

5. Adapting for a Sports Tech Sales Team

The tool was built for a partnership BD motion but is fully general. Here is how a sports tech company would adapt the key components:

Target types	Trade associations, civic orgs, fraternal groups	Pro teams, college athletic depts, leagues, governing bodies, esports orgs, event companies
Scoring signals	Member network, runs events, SMERF category, room block needs	Operational complexity, tech stack gaps, event volume, team/venue count, budget signals
Value props	Hotel savings, referral revenue, booking platform	Your product's core value — customise per segment (e.g. ops efficiency, fan engagement, data insights)
Pitch angles	Revenue share, conference travel offer	ROI angle, competitive displacement, peer reference ('X team uses us')
Email tone	Partnership-led, relationship-first	Sales-led — problem-aware, demo-focused, ROI-driven
Segment focus (style profile)	SMERF (Social, Military, Educational, Religious, Fraternal)	Your ICP — e.g. 'mid-market pro sports orgs, 50–500 staff, VP Ops/RevOps buyer'

Example: sports tech research prompt

Replace the partner framework section of the prompt with something like this:

```
Research "{company}" as a potential customer for [Your Product]. [Your Product] helps sports organisations automate [core use case]. Score this org on: 1. OPERATIONAL COMPLEXITY – multi-team, multi-venue, or high event volume 2. TECHNOLOGY GAP – manual processes your product replaces 3. BUYING SIGNALS – hiring ops/tech roles, recent funding, new venue/league 4. ICP FIT – pro sports, college athletics, esports, governing bodies Return value props, pitch angles and talking points tailored to their specific operational challenges.
```

6. Optional Integrations

ZoomInfo (recommended — add when ready)

ZoomInfo enriches the prospect brief with verified data: employee count, revenue range, HQ location, number of locations, and a company description. The tool is built to call ZoomInfo automatically alongside the Anthropic research call — it runs in parallel and merges the data into the brief.

How to add ZoomInfo

1. Get access to ZoomInfo's MCP (Model Context Protocol) server — contact your ZoomInfo account rep.
2. Add two environment variables to Vercel: ZOOMINFO_MCP_URL and ZOOMINFO_MCP_API_KEY.
3. The `enrichZoomInfo()` function in `app/api/partner-research/route.ts` activates automatically when these variables are present.
4. No code changes needed — the integration is already built in and just needs the credentials.

Supabase (optional — for team-wide data sharing)

By default, briefs and style profiles are stored in the browser (`localStorage`). This means each person's data stays on their own machine. If you want team-wide shared reports, a central activity log, or cross-rep pipeline visibility, add Supabase (a free-tier Postgres database).

- Create a free project at supabase.com
- Add `NEXT_PUBLIC_SUPABASE_URL` and `NEXT_PUBLIC_SUPABASE_ANON_KEY` to Vercel environment variables
- Create the reports table using the schema in `/supabase/schema.sql` (in the repo)
- The report-saving and loading logic already calls Supabase when these env vars are present

Crossbeam (optional — partner overlap signals)

Crossbeam shows you which of your prospects are already in a partner's customer base — enabling warm introductions. Useful for sales teams with an active partner ecosystem.

- Add `CROSSBEAM_API_KEY` to Vercel environment variables
- The `checkCrossbeam()` function in the research route activates automatically
- Overlap signals appear in the brief under 'Warm Paths' with the partner name and overlap type

7. Using Claude Cowork to Build and Maintain It

Claude Cowork is the fastest way to build and iterate on this tool without deep coding experience. Here is how to use it effectively at each stage:

Initial build — prompting Claude Cowork

When you first set up the project and open it in Claude Cowork, describe what you want in plain language:

```
"I have a Next.js project on Vercel that I'm building into a sales intelligence tool. I want it to: (1) let me enter a prospect org name and generate a full brief with a fit score and pitch angles; (2) show saved briefs in a sidebar; (3) export as PDF. My ICP is mid-market sports tech companies. Here is my scoring framework: [paste it]. Please build the research API route and the UI for the brief display."
```

Iterating on the scoring

After running a few briefs, you will notice orgs that score higher or lower than expected. Tell Claude Cowork:

```
"The brief for [Org X] scored 45 but I think they are a much stronger fit than that. They run 12 events per year and have a decentralised booking problem. Can you adjust the scoring prompt to weight recurring events more heavily?"
```

Adding new features

Claude Cowork can build net-new features on request. Examples of things you might add later:

- A CRM sync button that pushes brief data to Salesforce or HubSpot
- A competitor overlap section (flag if the org is a known customer of a competitor)
- A contact finder panel (pull the right decision-maker name + email for a given org)
- A weekly digest email summarising follow-ups due this week
- A team view showing all reps' pipelines in one dashboard

Maintaining the codebase

You do not need to understand all the code to maintain this tool. If something breaks, open the project folder in Claude Cowork and describe what is happening. Claude can read the error, find the relevant file, and fix it. The most common issues are:

- Vercel timeout errors — Claude can switch the affected route to Edge Runtime or optimise the prompt
 - Anthropic rate limit errors — Claude can add retry logic or reduce token usage
 - UI changes — describe what you want visually and Claude will update the React components
 - New environment variables needed — Claude will tell you what to add to Vercel's settings
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8. Environment Variables Reference

Add these in Vercel: Project Settings → Environment Variables. Never put them in your code or commit them to GitHub.

ANTHROPIC_API_KEY	Required	console.anthropic.com → API Keys
NEXT_PUBLIC_SUPABASE_URL	Optional	supabase.com → Project Settings → API
NEXT_PUBLIC_SUPABASE_ANON_KEY	Optional	supabase.com → Project Settings → API
ZOOMINFO_MCP_URL	Optional	Your ZoomInfo account rep
ZOOMINFO_MCP_API_KEY	Optional	Your ZoomInfo account rep
CROSSBEAM_API_KEY	Optional	crossbeam.com → Settings → API

9. Key Files in the Codebase

When asking Claude Cowork to make changes, reference these files by name to give it the right context immediately:

<code>app/page.tsx</code>	The entire frontend — UI, navigation, state, all React components. This is the main file.
<code>app/api/partner-research/route.ts</code>	The research engine. Contains the scoring prompt, Anthropic API call, ZoomInfo enrichment and Crossbeam signals. Customise this to change scoring.
<code>app/api/export-brief/route.ts</code>	Generates the downloadable Word document version of a brief (optional — PDF export is client-side).
<code>app/api/claude/route.ts</code>	Handles email generation, style analysis and other Claude calls from the outreach workflow.
<code>app/api/research/route.ts</code>	Handles the contact discovery / prospect search functionality.
<code>.env.local</code>	Local environment variables (never commit this file). Mirror these in Vercel's environment variable settings.

`next.config.ts`

Next.js configuration. Rarely needs changing.

10. Tips & Common Mistakes

Things that will save you time

- Spend the most time on the scoring prompt — it is the brain of the tool. Iterate on it with real orgs until scores feel right.
- Set up your style profile before generating any outreach emails. The more example emails you give it, the better the output.
- Use the Meeting Notes field on the research panel — even two sentences about the org changes the brief significantly.
- Save briefs as you go. The saved briefs panel means you never have to re-research the same org.
- The PDF export is entirely client-side — it opens in a new tab and uses the browser's print dialog. Tell your team to use 'Save as PDF' not 'Print'.

Common mistakes to avoid

- Do not add web search to the research API call on Vercel Hobby — it causes 504 timeouts. Claude's training knowledge is sufficient for most orgs. Upgrade to Pro if you need live web data.
- Do not store your Anthropic API key in the codebase or commit it to GitHub. Always use environment variables.
- Do not make the scoring prompt too long — it increases token usage and can hit rate limits. Keep it focused on your top 6–8 signals.
- Do not try to use the docx export route on Edge Runtime — it needs Node.js APIs. Keep it as a standard serverless route.
- Do not use the Hobby plan's localStorage for team-wide data. If you want shared pipeline visibility, add Supabase from the start.

Approximate cost guide

- Anthropic API (claude-haiku-4-5): ~\$0.003 per brief generated. 500 briefs/month ≈ \$1.50.
- Email generation: ~\$0.001 per email. 1,000 emails/month ≈ \$1.00.
- Vercel Hobby: Free. Vercel Pro: \$20/month (removes function timeout limits).
- ZoomInfo: Enterprise pricing — contact their sales team. Adds verified data quality.
- Supabase: Free tier covers most small teams. Pro at \$25/month for larger usage.
- Total for a small team (3–5 reps): typically \$5–30/month all-in.

11. Sample Prompt to Get Started with Claude Cowork

Copy and paste this into Claude Cowork (with your project folder selected) to kick off the build. Replace the bracketed sections with your specifics.

I want to build a sales intelligence and meeting prep tool using Next.js on Vercel, powered by the Anthropic API. I have an API key ready. The tool should:

1. Let me enter an org name and generate a full prospect brief with:
 - Company snapshot (industry, size, description, website)
 - Fit score (0-100) based on my ICP criteria
 - Fit signals (why they scored that way)
 - Tailored value props for my product
 - 2-3 pitch angles with opening lines
 - Key talking points
 - Recent news / timing signals
2. Save researched orgs in a sidebar so I can pull them up before calls
3. Export any brief as a clean PDF to share with my team
4. Have a sidebar navigation with: Outreach, Reports, Research & Pitch, Settings

My company: [Your company name]
My product: [What it does in one sentence]
My ICP: [Describe your ideal customer – industry, size, role, pain points]
Scoring criteria (0-100 fit score): [Paste your scoring criteria here – see Section 4 of this guide]
Please start by building the research API route and the brief display UI. Use Edge Runtime for the research route to avoid Vercel timeout issues.

You have everything you need to build this.

The tool was built iteratively over several weeks using Claude Cowork — no dedicated engineering team, no prior Next.js experience required. Start with the research brief, get it working, then layer in outreach, reporting, and integrations one at a time.

- Questions about the build? Open the project in Claude Cowork and ask — it has the full context.
- Something is broken? Describe the error to Claude Cowork with the file name and it will fix it.
- Want to add a feature? Describe it in plain language — Claude Cowork will build it.